

# AI Engineering Lab

## Answer Key — Quiz Form A

The correct option of each item is set in bold. Each item also shows its blueprint cell and the misconception its distractors target.

Internal document — do not hand out to students.

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### Part C Knowledge questions

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20 questions, each with exactly one best answer. Tick one box per question. If you do not know, please tick “I do not know” rather than guessing — an honest blank is more useful to us than a lucky guess, and none of this affects your grade. Plan for about 15 minutes.

1. The LLM in an agentic system is usually described as *stateless across calls*. What does that mean?
  - (a) Its weights are frozen, so it cannot be adapted to new data.
  - (b) It retains nothing from one call to the next, so everything it needs must be supplied in the current request.**
  - (c) The provider stores the conversation and reloads it automatically for each call.
  - (d) It returns the same output for the same input, because it holds no random state.

☐ I do not know.

[topic: T1 — LLM role & statelessness | level: remember | difficulty: easy | blueprint cell: A01]

**Solution:** *Statelessness is about the call boundary, not about training. (a) confuses statelessness with frozen weights; (c) is the opposite of statelessness; (d) confuses it with determinism.*

2. Why does a harness try to keep the context small even in runs that are nowhere near filling the context window?
  - (a) Because model performance degrades as the window fills, so the goal is the smallest set of high-signal tokens that still produces the desired behaviour.**
  - (b) Because tokens that are not needed in the current step are discarded by the model anyway.
  - (c) Because the context window is shared between all agents using the same endpoint.
  - (d) Because a shorter context guarantees that the loop terminates in fewer iterations.

☐ I do not know.

[topic: T2 — Context engineering | level: understand | difficulty: medium | blueprint cell: A02]

**Solution:** *This is context rot: the constraint is degradation, not the hard limit. Distractors treat the window as a hard limit only, as shared state, or as a loop bound.*

3. A run has grown so long that the context no longer fits. The team wants to keep the agent coherent while spending fewer tokens. Which pair of measures matches this goal?
- (a) Raise the temperature so the model relies less on the context, and shorten the system prompt.
  - (b) Split the context evenly across several LLM calls and merge the answers afterwards.
  - (c) **Summarise the older turns, and move state that is not needed right now to external storage that can be read back on demand.**
  - (d) Drop the tool schemas from the request, since the model has already used those tools once.
- ☐ I do not know.

[topic: T2 — Context engineering | level: apply-analyse | difficulty: hard | blueprint cell: A03]

**Solution:** *Compaction plus offloading. (d) is attractive but breaks tool calling, because schemas are not remembered across calls — it also tests T1.*

4. Why does the loop need guard conditions such as an iteration cap or a token budget?
- (a) **Because the LLM may return neither a final answer nor a valid action, and without a bound the loop would never terminate.**
  - (b) Because the model provider charges a penalty for requests above a fixed rate.
  - (c) Because every iteration adds a tool to the registry, which would eventually exhaust the schema limit.
  - (d) Because the user must be given a chance to intervene after a fixed number of steps.
- ☐ I do not know.

[topic: T3 — Loop and control flow | level: understand | difficulty: medium | blueprint cell: A04]

**Solution:** *Guards guarantee termination. (d) confuses guard conditions with permission gates.*

5. A team extends their loop so that the agent first writes a plan for the task and then executes it. What must the loop now do that a plain ReAct loop does not?
- (a) Call the LLM twice per iteration, once for reasoning and once for acting.
  - (b) Persist the plan to long-term memory before any tool may be called.
  - (c) Disable its iteration cap, because the plan already bounds the number of steps.
  - (d) **Contrast the progress made so far against that initial plan on each pass through the loop.**
- ☐ I do not know.

[topic: T3 — Loop and control flow | level: apply-analyse | difficulty: hard | blueprint cell: A05]

**Solution:** *Up-front planning adds a comparison against the plan. (c) is the dangerous misconception: a plan is not a termination guarantee.*

6. Which of the following best describes a *tool* in an agentic system?

- (a) A store of facts that survives from one agent run to the next.
- (b) An interface with a defined schema through which the agent retrieves information or acts on an environment.**
- (c) A prompt template that instructs the model how to solve a class of tasks.
- (d) The component that decides which action the agent should take next.

☐ I do not know.

[topic: T4 — Tools and tool design | level: remember | difficulty: easy | blueprint cell: A06]

**Solution:** The distractors are the neighbouring components: long-term memory (a), a skill (c), the LLM/loop (d).

7. A `read_file` tool returns an empty string both when the file is empty and when the path does not exist. What is the most likely consequence for the agent?

- (a) It retries the call with the same arguments until the loop's iteration cap is reached.
- (b) The tool registry rejects the call, because an empty result violates the tool's schema.
- (c) It cannot tell the two situations apart and reasons on from a wrong observation, because the failure was never communicated to it.**
- (d) The loop enters its terminal state, because an empty observation carries no information.

☐ I do not know.

[topic: T4 — Tools and tool design | level: apply-analyse | difficulty: hard | blueprint cell: A07]

**Solution:** Tool failures must be communicated upstream. Silent, ambiguous success values are worse than errors because the agent has no signal that anything went wrong.

8. What is the Model Context Protocol (MCP)?

- (a) A language model specialised in managing an agent's context window.
- (b) A protocol through which an agent connects to servers that expose tools and other capabilities.**
- (c) A file format for storing an agent's long-term memory between runs.
- (d) A specification of how much context a model must be able to process.

☐ I do not know.

[topic: T5 — MCP | level: remember | difficulty: easy | blueprint cell: A08]

**Solution:** (a) and (d) target the widespread "MCP is a model / a context size standard" confusion invited by the name.

9. A requirement says the tools of an MCP server must be made available to the model *without hard-coding individual tools*. What does the harness have to do?

- (a) Generate a Python wrapper function per tool before the application is built.
- (b) Keep a configuration file that lists every tool name the server is known to provide.
- (c) Ask the model which tools it thinks exist and validate its guesses against the server.
- (d) **Query the server for its tool list at runtime and pass the returned schemas on to the model.**

☐ *I do not know.*

[topic: T5 — MCP | level: apply-analyse | difficulty: medium | blueprint cell: A09]

**Solution:** *Runtime discovery. (a) and (b) are both hard-coding, just at different times.*

10. Why does a harness usually *not* keep all available skills in the context at all times?

- (a) Because skills may contradict one another if more than one is loaded at a time.
- (b) Because the model can execute only one skill per run.
- (c) Because a skill has to be validated by a hook before it may enter the context.
- (d) **Because the context should carry only what the current task needs; a skill is disclosed when the task calls for it.**

☐ *I do not know.*

[topic: T6 — Skills | level: understand | difficulty: medium | blueprint cell: A10]

**Solution:** *Just-in-time disclosure follows from the context budget, not from conflicts between skills.*

11. What characterises long-term memory, as opposed to context?

- (a) It is the part of the context window that is never summarised away.
- (b) It is the conversation history of the current run, kept in full.
- (c) **It lives outside the context window, persists across runs, and is consulted on demand.**
- (d) It is the larger context window offered by models that support one.

☐ *I do not know.*

[topic: T7 — Long-term memory | level: remember | difficulty: easy | blueprint cell: A11]

**Solution:** *(d) is the “long-term memory is just a bigger window” misconception.*

12. What is a hook in an agent harness?

- (a) A tool the agent may call to inspect its own state.
- (b) A custom function that runs at a defined point in the agent's lifecycle, such as before a tool call.**
- (c) A connection to an external server from which tools are discovered.
- (d) A rule that decides whether the agent needs the user's approval.

☐ I do not know.

[topic: T8 — Hooks | level: remember | difficulty: easy | blueprint cell: A12]

**Solution:** *Distractors are, in order, a tool, an MCP connection and a permission policy.*

13. A team wants to block any tool call whose arguments point outside the sandbox root, *before* it is executed. Which mechanism is designed for that?

- (a) A pre-use hook that inspects the arguments and can reject the call.**
- (b) A post-use hook that reports the violation once the tool has run.
- (c) A sub-agent that reviews the main agent's proposed actions.
- (d) A longer system prompt instructing the model never to leave the sandbox.

☐ I do not know.

[topic: T8 — Hooks | level: apply-analyse | difficulty: medium | blueprint cell: A13]

**Solution:** *(d) is the "prompt instead of enforce" misconception; (b) has the right mechanism but the wrong lifecycle point.*

14. In a tiered permission model, which operation belongs in the tier that requires mandatory confirmation by the user?

- (a) Listing the files in the agent's working directory.
- (b) Running the project's unit tests inside the container.
- (c) Reading a configuration file in the sandbox.
- (d) Opening a pull request against a public repository.**

☐ I do not know.

[topic: T9 — Permission management | level: understand | difficulty: easy | blueprint cell: A14]

**Solution:** *The criterion is whether the consequences are irreversible or leave the sandbox, not how complicated the operation is.*

15. A permission policy supports the outcomes *allow*, *deny* and *require user confirmation*. Why is the third outcome needed rather than deciding every case in advance?

- (a) Because the policy engine cannot evaluate its rules fast enough to keep the loop responsive.
- (b) Because the model cannot produce arguments that a static rule would be able to check.
- (c) **Because for some actions the consequences depend on the situation, so the judgement has to be handed to a human at the moment it arises.**
- (d) Because denying an action would end the run, which the harness has to avoid.

☐ *I do not know.*

[topic: T9 — Permission management | level: understand | difficulty: medium | blueprint cell: A15]

**Solution:** *Human-in-the-loop gates exist for context-dependent risk. (d) misstates what a denial does.*

16. Besides containment, what second benefit does a stable sandbox provide?

- (a) Lower latency, because the sandboxed environment sits closer to the model endpoint.
- (b) A guarantee that the agent cannot loop indefinitely.
- (c) Automatic approval of low-risk actions without asking the user.
- (d) **Reproducibility: the same command, seed and dependency set can be replayed under comparable conditions, so a failure reflects a real defect rather than environment drift.**

☐ *I do not know.*

[topic: T10 — Sandboxing | level: understand | difficulty: medium | blueprint cell: A16]

**Solution:** *Reproducibility is the second stated purpose of containerisation in this lab. (b) and (c) attribute loop control and permission decisions to the sandbox.*

17. A main agent delegates a subtask to a specialised sub-agent. What does the main agent get back, and why does that matter?

- (a) **A condensed result, so that the sub-agent's detailed intermediate work never enters the main agent's context.**
- (b) The sub-agent's full transcript, so that the main agent can verify every step it took.
- (c) The sub-agent's tool schemas, so that the main agent could repeat the work itself.
- (d) Nothing — the sub-agent writes its result to long-term memory, which the main agent reads later.

☐ *I do not know.*

[topic: T11 — Sub-agents and A2A | level: understand | difficulty: medium | blueprint cell: A17]

**Solution:** *The point of delegation is context isolation; (b) would defeat it entirely.*

18. Which set of measurements is most relevant for monitoring an agentic system in operation?

- (a) **Run throughput and latency, success and failure rates, model and token usage, tool usage, and permission decisions.**
- (b) CPU temperature, disk fragmentation, and the number of open file descriptors.
- (c) The number of lines of code in the harness and its test coverage.
- (d) The model's parameter count, context window size, and training cut-off date.

☐ *I do not know.*

[topic: T12 — Observability and telemetry | level: understand | difficulty: easy | blueprint cell: A18]

**Solution:** (c) measures the codebase and (d) the model, neither of which changes while the system runs.

19. Your dashboard shows that the tool failure rate rose from 2% to 20% this morning. What do you need *in addition* to the aggregate metrics in order to find the cause?

- (a) A longer retention period for the same aggregate counters.
- (b) A second dashboard showing the same metrics with a finer colour scale.
- (c) **Structured logs or traces that let you inspect what happened inside individual runs.**
- (d) More frequent scraping, so that the same counters are sampled at a finer time resolution.

☐ *I do not know.*

[topic: T12 — Observability and telemetry | level: apply-analyse | difficulty: medium | blueprint cell: A19]

**Solution:** Aggregates detect; traces diagnose. (d) improves resolution of the same aggregate signal and still cannot explain a single run.

20. An end-to-end test instructs the agent to create `output/answer.md` containing the text `forty-two`. Which assertion is robust against the agent's non-determinism?

- (a) The agent called `create_folder` before `create_file`.
- (b) **The file exists at that path and its contents are exactly the expected text.**
- (c) The agent needed exactly two tool calls to complete the task.
- (d) The agent's final message contains the sentence "I have created the file."

☐ *I do not know.*

[topic: T13 — Quality assurance | level: understand | difficulty: medium | blueprint cell: A20]

**Solution:** Assert on resulting state, not on the path taken or the wording chosen — both vary between runs even when the outcome is correct.